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MemoryBank: Enhancing Large Language Models with Long-Term Memory

Wanjun Zhong[1], **Lianghong Guo**[1], **Qiqi Gao**[2], **He Ye**[3], **Yanlin Wang**[1]
[1] Sun Yat-Sen University [2] Harbin Institute of Technology
[3] KTH Royal Institute of Technology
{zhongwj25@mail2, wangylin36@mail}.sysu.edu.com
2231612405@qq.com, 18b903026@stu.hit.edu.cn
heye@kth.se

Abstract

Revolutionary advancements in Large Language Models (LLMs) have drastically reshaped our interactions with artificial intelligence (AI) systems, showcasing impressive performance across an extensive array of tasks. Despite this, a notable hindrance remains—the deficiency of a long-term memory mechanism within these models. This shortfall becomes increasingly evident in situations demanding sustained interaction, such as personal companion systems, psychological counseling, and secretarial assistance. Recognizing the necessity for long-term memory, we propose MemoryBank, a novel memory mechanism tailored for LLMs. MemoryBank enables the models to summon relevant memories, continually evolve through continuous memory updates, comprehend, and adapt to a user's personality over time by synthesizing information from previous interactions. To mimic anthropomorphic behaviors and selectively preserve memory, MemoryBank incorporates a memory updating mechanism, inspired by the Ebbinghaus Forgetting Curve theory. This mechanism permits the AI to forget and reinforce memory based on time elapsed and the relative significance of the memory, thereby offering a more human-like memory mechanism and enriched user experience. MemoryBank is versatile in accommodating both closed-source models like ChatGPT and open-source models such as ChatGLM. To validate MemoryBank's effectiveness, we exemplify its application through the creation of an LLM-based chatbot named SiliconFriend in a long-term AI Companion scenario. Further tuned with psychological dialog data, SiliconFriend displays heightened empathy and discernment in its interactions. Experiment involves both qualitative analysis with real-world user dialogs and quantitative analysis with simulated dialogs. In the latter, ChatGPT acts as multiple users with diverse characteristics and generates long-term dialog contexts covering a wide array of topics. The results of our analysis reveal that SiliconFriend, equipped with MemoryBank, exhibits a strong capability for long-term companionship as it can provide emphatic response, recall relevant memories and understand user personality. This underscores the effectiveness of MemoryBank[1].

1 Introduction

The advent of Large Language Models (LLMs) such as ChatGPT (OpenAI, 2022) and GPT-4 (OpenAI, 2023) has led to increasing influence across various sectors, from education and healthcare to customer service and entertainment. These powerful AI systems have demonstrated a remarkable ability to

[1]The materials are released in `https://github.com/zhongwanjun/MemoryBank-SiliconFriend`. The corresponding author is Yanlin Wang.

understand and generate human-like responses. Despite the remarkable capabilities of LLMs, a key limitation is their lack of long-term memory, an essential aspect of human-like communication, particularly noticeable in scenarios requiring sustained interactions like personal companionship, psychological counseling, and secretarial tasks. Long-term memory in AI is vital to maintain contextual understanding, ensure meaningful interactions and understand user behaviors over time. For instance, personal AI companions need to recall past conversations for rapport building. In psychological counseling, an AI can provide more effective support with knowledge of the user's history and past emotional states. Similarly, secretarial AI requires memory for task management and preference recognition. The absence of long-term memory in LLMs hinders their performance and user experience. Therefore, it is essential to develop AI systems with improved memory capabilities for a more seamless and personalized interaction.

Therefore, we introduce MemoryBank, a novel mechanism designed to provide LLMs with the ability to retain long-term memory and draw user portraits. MemoryBank enables LLMs to recall historical interactions, continually evolve their understanding of context, and adapt to a user's personality based on past interactions, thereby enhancing their performance in long-term interaction scenarios. Inspired by the Ebbinghaus Forgetting Curve theory, a well-established psychological principle that describes how the strength of memory decreases over time, MemoryBank further incorporates a dynamic memory mechanism closely mirroring human cognitive process. This mechanism empowers the AI to remember, selectively forget, and strengthen memories based on time elapsed, offering more natural and engaging user experience. Specifically, MemoryBank is built on a memory storage with memory retrieval and updating mechanism, and ability to summarize past events and users' personality.

MemoryBank is versatile as it can accommodate both closed-source LLMs like ChatGPT and open-source LLMs like ChatGLM (Zeng et al., 2022) or BELLE (Yunjie Ji & Li, 2023).

To exemplify the practical implications of MemoryBank, we develop SiliconFriend, an LLM-based AI Companion chatbot integrated with this innovative memory mechanism. SiliconFriend is designed to retain and reference past interactions, reinforcing the transformative influence of MemoryBank in crafting a more personable AI companion. A distinctive features of SiliconFriend is its tuning with 38k psychological conversations, collected from various online sources, which enables it to exhibit empathy, carefulness, and provide useful guidance, making it adept at handling emotionally charged dialogues. Moreover, one of the standout capabilities of SiliconFriend is to understand a user's personality by summarizing from past interactions, which empowers it to tailor responses to the user's individual traits, thereby enhancing user experience. Additionally, SiliconFriend supports bilingual functionality, catering to users who communicate in English and Chinese. This multi-language support broadens its accessibility and usability across different user groups. SiliconFriend is implemented with two open-source models, ChatGLM and BELLE, along with one closed-source model, ChatGPT, showcasing the versatility of MemoryBank in accommodating different LLMs.

To evaluate the effectiveness of MemoryBank, we conduct evaluations covering both qualitative and quantitative analyses, where the former involves real-world user dialogs and the latter employs simulated dialogs. For the quantitative analysis, we create a memory storage consisting of 10 days of conversations encompassing a diverse range of topics. These conversations involve 15 virtual users with diverse personalities, for which ChatGPT plays the role of users and generates dialog contexts according to their personalities. Based on this memory storage, we design 194 probing questions to assess whether the model could successfully recall pertinent memories and provide appropriate responses. Experiment results showcase the capabilities of SiliconFriend in memory recall, provision of empathetic companionship, and understanding of user portraits. These findings corroborate the potential of MemoryBank to significantly improve the performance of LLMs in long-term interaction scenarios. In this paper, we summarize the key contributions as follows:

- We introduce MemoryBank, a novel human-like long-term memory mechanism, which enables LLMs to store, recall, update memory, and draw user portrait.
- We demonstrate the practical applicability of MemoryBank through SiliconFriend, an LLM-based AI companion equipped with MemoryBank and tuned with psychological dialogs. It can recall past memories, provide empathetic companionship, and understand user behaviors.
- We show the generalizability of MemoryBank in three key aspects: (1) Accommodation of both open-source and closed-source LLMs; (2) Bilingual ability in both Chinese and English; (3) Applicability with and without memory forgetting mechanism.

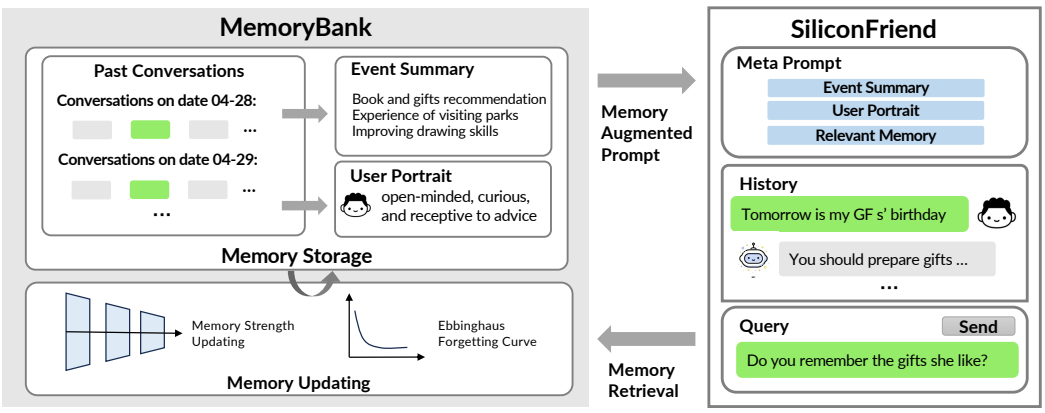

Figure 1: Overview of MemoryBank. The memory storage (§ 2.1) stores past conversations, summarized events and user portraits, while the memory updating mechanism (§ 2.3) updates the memory storage. Memory retrieval (§ 2.2) recall relevant memory. SiliconFriend (§ 3) serves as an LLM-based AI companion augmented with MemoryBank.

2 MemoryBank: A Novel Memory Mechanism Tailored for LLMs

In this section, we provide a detailed description of MemoryBank, our novel memory mechanism designed for LLMs. As shown in Fig. 1, MemoryBank is a unified mechanism structured around three central pillars: (1) a memory storage (§ 2.1) serving as the primary data repository, (2) a memory retriever (§ 2.2) for context-specific memory recollection, and (3) a memory updater (§ 2.3) drawing inspiration from the Ebbinghaus Forgetting Curve theory, a time-tested psychological principle pertaining to memory retention and forgetting.

2.1 Memory Storage: The Warehouse of MemoryBank

Memory storage, the warehouse of MemoryBank, is a robust data repository holding a meticulous array of information. As shown in Fig. 1, it stores daily conversations records, summaries of past events, and evolving assessments of user personalities, thereby constructing a dynamic and multi-layered memory landscape.

In-Depth Memory Storage: MemoryBank's storage system captures the richness of AI-user interactions by recording multi-turn conversations in a detailed, chronological fashion. Each piece of dialogue is stored with timestamps, creating an ordered narrative of past interactions. This detailed record not only aids in precise memory retrieval but also facilitates the memory updating process afterwards, offering a detailed index of conversational history.

Hierarchical Event Summary: Reflecting the intricacies of human memory, MemoryBank goes beyond mere detailed storage. It processes and distills conversations into a high-level summary of daily events, much like how humans remember key aspects of their experiences. We condense verbose dialogues into a concise daily event summary, which is further synthesized into a global summary. This process results in a hierarchical memory structure, providing a bird's eye view of past interactions and significant events. Specifically, taken previous daily conversations or daily events as input, we ask the LLMs to summarize daily events or global events with the prompt "*Summarize the events and key information in the content* `[dialog/events]`".

Dynamic Personality Understanding: MemoryBank focuses on user personality understanding. It continuously assesses and updates these understandings with the long-term interactions and creates daily personality insights. These insights are further aggregated to form a global understanding of the user's personality. This multi-tiered approach results in an AI companion that learns, adapts, and tailors its responses to the unique traits of each user, enhancing user experience. Specially, taken the daily conversations or personality analysis, we ask the LLM to deduce with prompts: "*Based on the following dialogue, please summarize the user's personality traits and emotions.* `[dialog]`" or "*The*